Bonnyrigg Chess Engine-School Tournament Engine

By Danny Phan

GITHUB:HTTPS://GITHUB.COM/DANNYP42/BONNYRIGG-CHESS-ENGINE-SCHOOL-TOURNAMENT-ENGINE-BY-DANNY-PHAN.GIT

?Why this Chess Engine

Existing chess software is often complete

Difficult for students to understand

Need for an educational chess platform

Supports school tournaments

Users

Students

Teachers

Chess club members

Tournament organisers

Benefits

Encourages logical thinking

Supports learning programming concepts

Easy-to-use interface

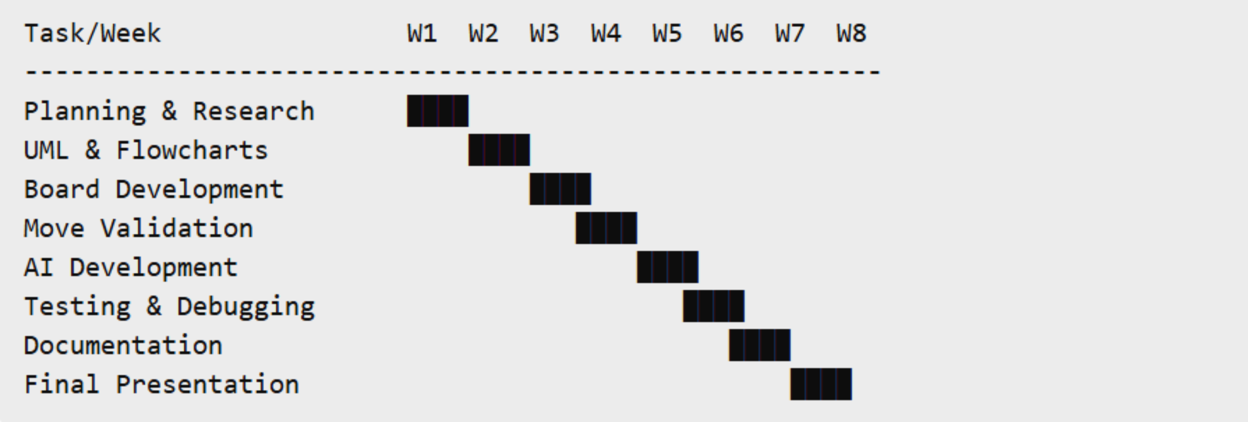
Methodology

Agile Development

- Incremental development
- Continuous testing
- Flexible improvements

Sprint Plan

- Board Setup
- Move Validation
- User Interface
- Testing & Refinement



Technologies Used

Programming Language

- Python

Tools

- VS Code
- GitHub
- Pygame
- Tkinter
- **?Why these tools**
- Simple syntax
- Easy debugging
- Strong documentation
- Excellent version control

System Architecture

Main Components

- GUI Layer
- Game Engine
- Move Validator
- Database System
- Export Module

```
┌───────────────────┐
│       User        │
└───────────────────┘
         │
         │
         ▼
┌───────────────────┐
│   GUI Interface   │
│     (Pygame)      │
└───────────────────┘
         │
         │
         ▼
┌───────────────────┐
│    Game Engine    │
│  ChessEngine.py   │
└───────────────────┘
         │
         │
 ┌───────┼─────────┐
 ▼       ▼         ▼
┌────────────┐ ┌────────────┐ ┌────────────┐
│Move        │ │Game State  │ │Piece Logic │
│Validation  │ │Management  │ │Rules       │
└────────────┘ └────────────┘ └────────────┘
         │
         ▼
┌───────────────────┐
│ Save Data / Logs  │
└───────────────────┘
```

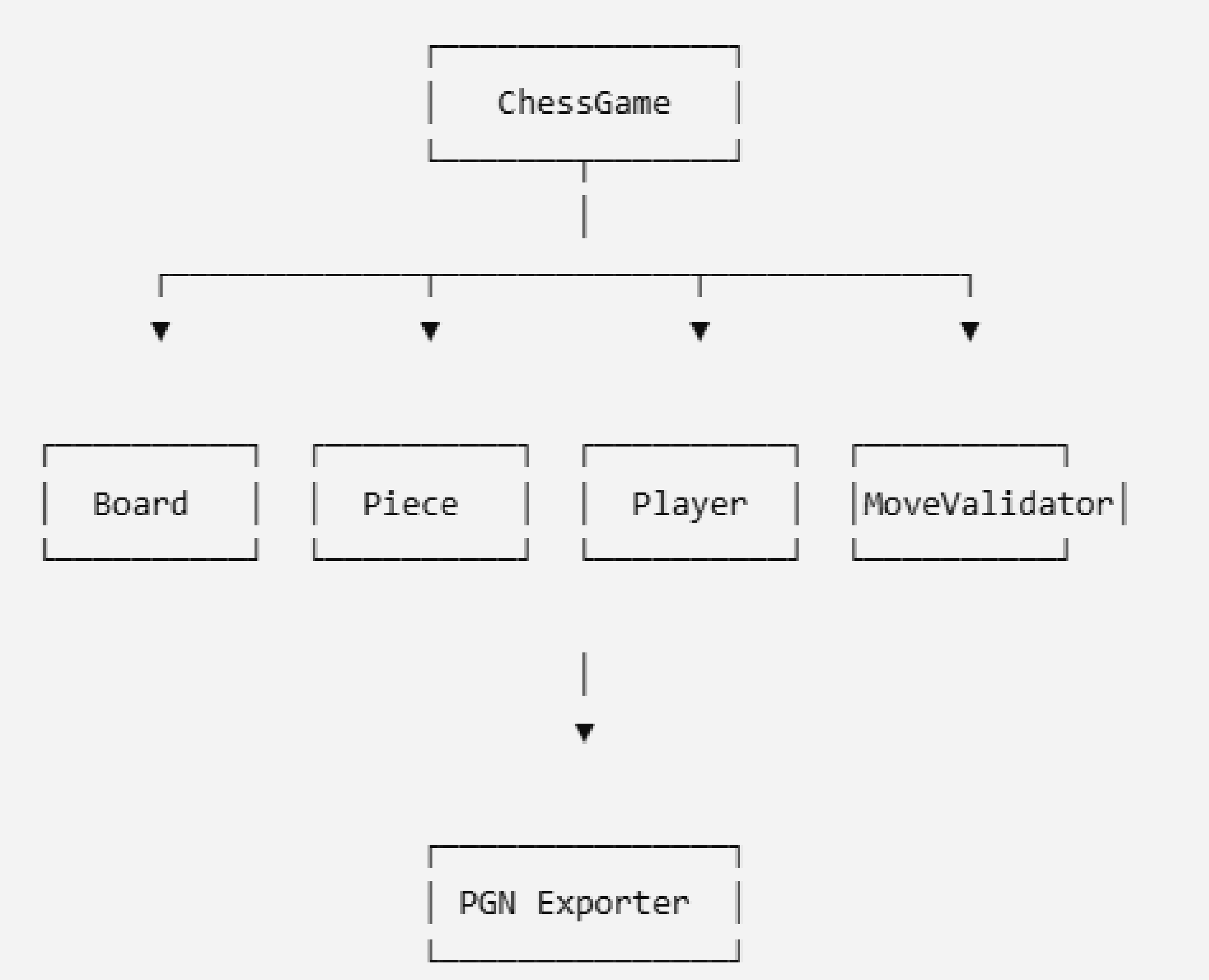
UML Design

Main Classes

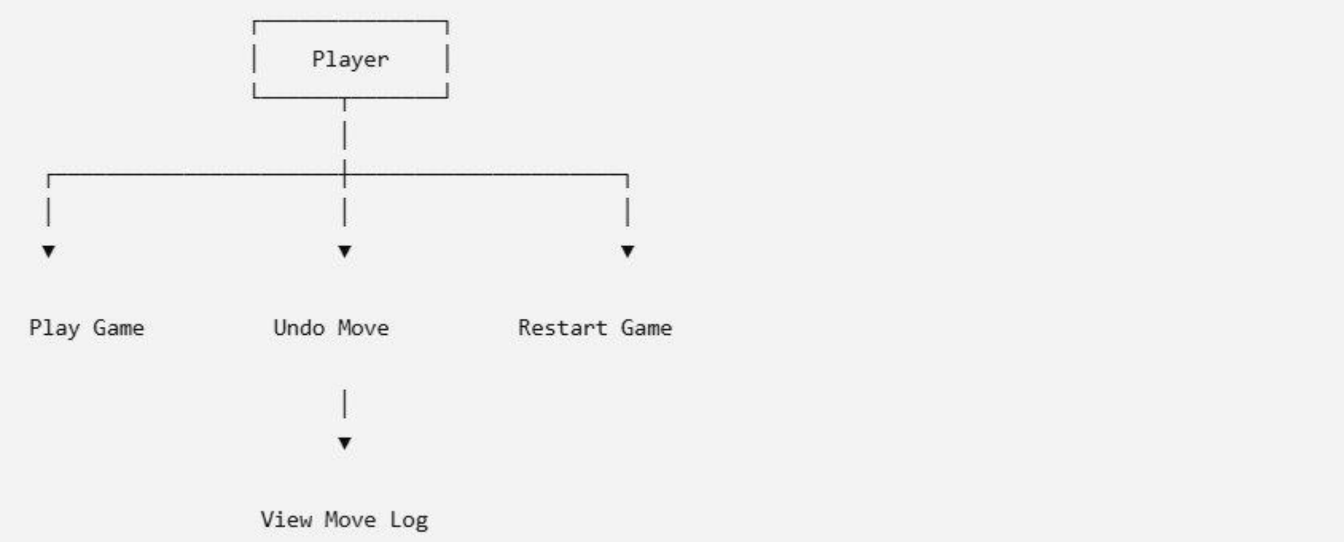
- ChessGame
- Board
- Piece
- MoveValidator
- Player
- PGNExporter

Key Relationships

- ChessGame controls gameplay
- Board stores piece locations
- MoveValidator checks legality



Case Diagram



Security

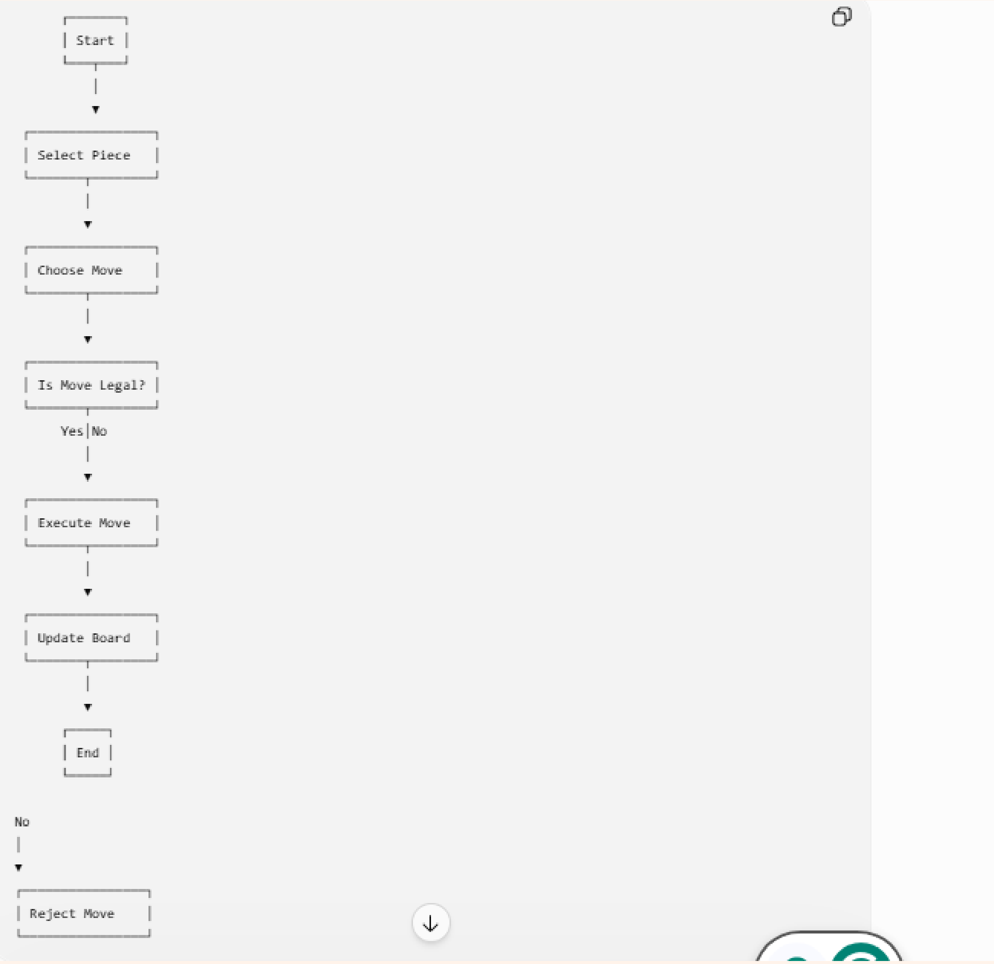
Security Features

- Input validation
- Exception handling
- Graceful failure
- Github backups

Future Improvements

- User authentication
- HTTPS support
- Data encryption

Validation flowchart



Testing

Unit Testing

- ✅ Legal Moves
- ✅ Checkmate Detection
- ✅ Castling
- ✅ Pawn Promotion

Accessibility Testing

- ✅ Keyboard Navigation
- ✅ Readable Interface

Edge Cases

- ✅ Stalemate
- ✅ Triple Repetition
- ✅ En Passant

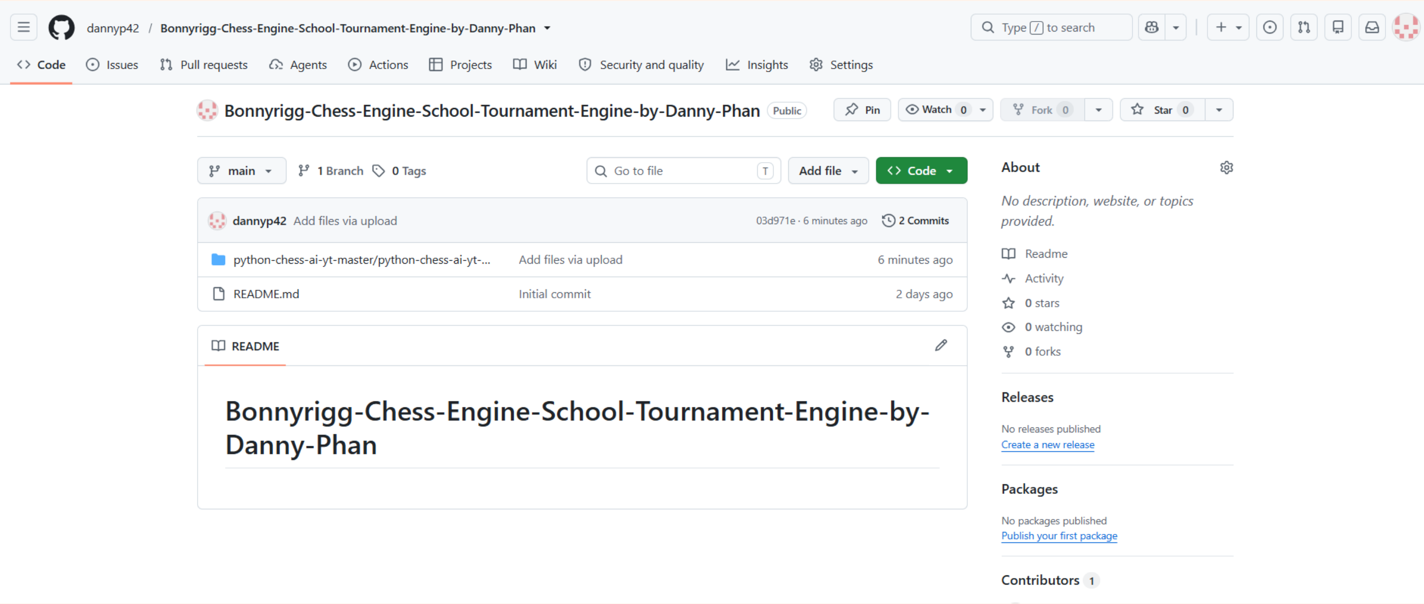
GitHub Management

GitHub Usage

- Source code storage
- Version control
- Backup system
- Collaboration support

Workflow

- Create Branch
- Develop Feature
- Test
- Commit
- Push
- Merge



Challenges

Major Challenges

- Move validation logic
- Special chess rules
- Debugging errors
- Understanding repository structure

Solutions

- Incremental testing
- Agile sprints
- Debugging tools

Future Improvements

Planned Features

- Online multiplayer
- Tournament mode
- Cloud save system
- Improved graphics
- Opening book support

Reflection

Skills Developed

- Python Programming
- Object-Oriented Programming
- Testing
- GitHub Version Control
- Documentation
- Problem Solving

Conclusion

- Met project requirements
- Functional chess engine created
- Demonstrated software engineering principles